

Functions



Function notation and evaluation

1 If $f(x) = 2x^2 - 3x + 1$, find $f(4)$.

2 If $g(x) = \frac{x+3}{x-1}$, find $g(-2)$.

Domain and range

3 Find the domain of $f(x) = \frac{1}{x-5}$.

4 Find the domain of $g(x) = \sqrt{x-3}$.

Composite functions

5 If $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $f(g(2))$.

6 If $f(x) = 3x - 2$ and $g(x) = x + 4$, find $(g \circ f)(x)$ as a simplified expression.

Inverse functions

7 Find the inverse of $f(x) = 3x - 6$.

8 If $f(5) = 12$, find $f^{-1}(12)$.

Transformations of functions

9 The graph of $y = f(x)$ is shifted 3 units right and 2 units up. Write the equation of the transformed function in terms of f .

10 Describe the transformation that takes $y = x^2$ to $y = -2(x+1)^2 + 5$.

Piecewise functions

11 A function is defined as $f(x) = x + 2$ for $x < 3$ and $f(x) = 2x - 1$ for $x \geq 3$. Find $f(1)$ and $f(5)$.

Interpreting function graphs and increasing/decreasing behavior

- 12** A function's graph rises from $x = 0$ to $x = 3$, reaches a peak at $x = 3$, then falls from $x = 3$ to $x = 6$. State the intervals where the function is increasing and decreasing, and identify the type of point at $x = 3$.

Modeling with function families

- 13** A company's profit is modeled by $P(x) = -2x^2 + 40x - 150$, where x is the number of units sold (in hundreds). Determine whether this is a linear, quadratic, or exponential model, and find the number of units that maximizes profit.

Finding function values from a graph or table

- 14** A table shows: $f(0) = 3$, $f(1) = 5$, $f(2) = 9$, $f(3) = 15$. Determine whether this could represent a linear or quadratic function, by examining the differences between consecutive values.

Zeros of a function and their graphical meaning

- 15** A function $f(x) = (x - 2)(x + 5)$ has zeros where it crosses the x -axis. Find the zeros, and state the y -intercept of the function.

Even and odd functions

- 16** Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, by evaluating $f(-x)$ and comparing to $f(x)$.
- 17** Determine whether $g(x) = x^3 - 2x$ is even, odd, or neither.

Modeling a real-world quantity with a function and evaluating it

- 18** The cost to rent a car is modeled by $C(d) = 35 + 0.20d$, where d is the number of miles driven. Find $C(150)$ and interpret the result in context.

Solving for an input given an output

19 If $f(x) = 2x - 7$ and $f(a) = 15$, find the value of a .

20 If $g(x) = x^2 - 4$ and $g(b) = 21$, find all possible values of b .